

Sustainable in-house PCB prototyping

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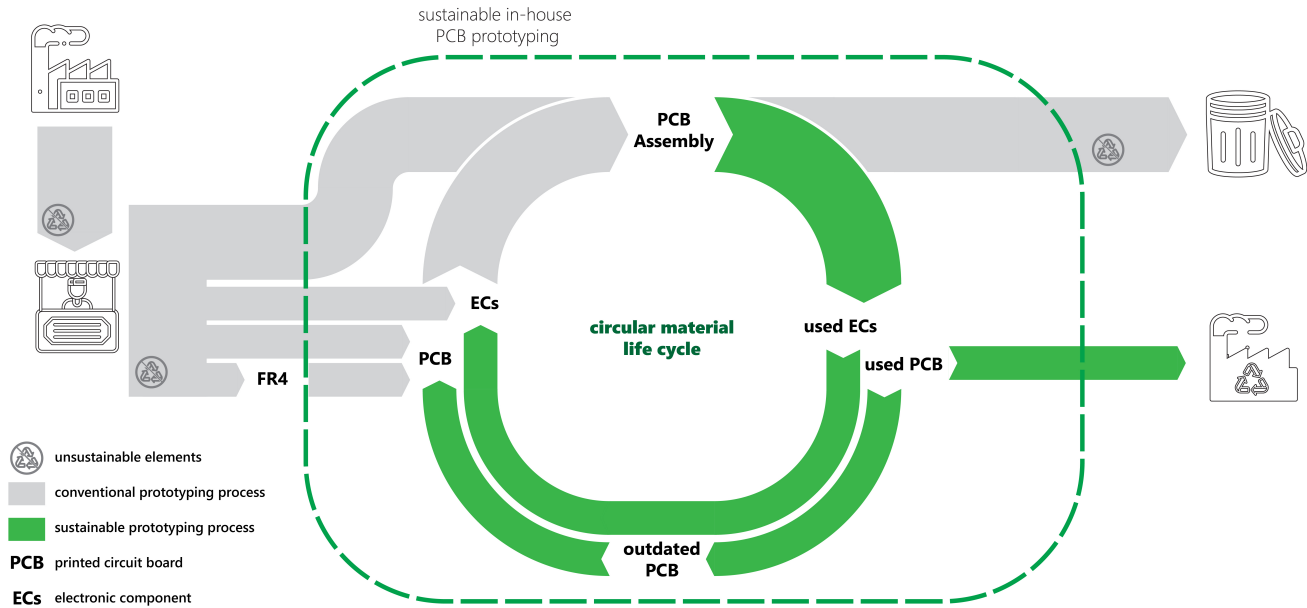


Figure 1: PCB prototyping process: black bounding box illustrates the conventional PCB prototyping process with all raw materials outsourced and most material scraped; green bounding box illustrates a sustainable PCB prototyping process which reuse and recycle PCB boards and components for latter iterations and projects, as well as help with external recycling by material separation.

ABSTRACT

Printed circuit boards (PCBs) are now ubiquitous in everyday objects. Despite advancements in mass production and the widespread availability of circuit design software, PCB manufacturing continues to present significant sustainability challenges due to the continual material sourcing and e-waste generation.

In this paper, I investigate the sustainability challenges and potential solutions in PCB production, utilizing in-house prototyping as a focal point. This approach is both accessible for investigation and allows the generalization of findings to broader contexts. I propose novel digital fabrication techniques to enhance material circulation between PCB projects and design iterations, thereby mitigating the environmental impact of electronics manufacturing. I introduce three research archetypes: SolderlessPCB, Fibercuit, and

PCB Renewal, each addressing sustainability challenges at different stages of PCB prototyping. These solutions possess the potential to be extrapolated to solve large-scale PCB manufacturing sustainability challenges, heralding a more sustainable future for electronics production.

CCS CONCEPTS

• **Social and professional topics** → Sustainability; • **Hardware** → Printed circuit boards; • **Human-centered computing** → Systems and tools for interaction design.

KEYWORDS

PCB prototyping, sustainability, reuse, renewal, prototyping, digital fabrication

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1 INTRODUCTION

Printed circuit boards (PCBs) are the foundation of electrical apparatus and electronic devices. With the explosive growth of technologies such as consumer electronics, the Internet of Things (IoT), renewable energy systems, electric vehicles, and medical devices, the demand for PCB design and prototyping has surged dramatically. However, this rapid increase in PCB development has contributed significantly to the growing issue of electronic waste (e-waste). According to the Global E-waste Monitor 2024, e-waste reached 62 million tonnes in 2022, an 82% increase from 2010, and is projected to rise to 82 million tonnes by 2030 (E-Waste Monitor). In addition to the e-waste problem, the material procurement phase of PCB production poses significant sustainability challenges. These include carbon emissions and energy consumption during the manufacture and transportation [6] of PCB substrate material and electronic components. Despite efforts to explore industrial-scale recycling methods and infrastructures to recover valuable materials from e-waste, these sustainability challenges highlight the urgent need for more sustainable and circular PCB-making processes.

To address these issues, I propose investigating the problem through the lens of in-house PCB prototyping. In recent decades, in-house PCB prototyping has become increasingly prevalent, driven by the maker movement and the democratization of printing and engraving technologies traditionally used for industrial-scale PCB production. Despite its smaller scale, the in-house PCB-making process closely mirrors large-scale manufacturing. Both utilize multi-layer laminated substrates consisting of fiberglass-polymer composites for dielectric layers and copper for conductive layers. Both processes adopt subtractive manufacturing techniques, such as machining or chemical etching, and use a universal set of electronic components for assembly. By focusing on in-house PCB prototyping, we can take advantage of its cost-effectiveness and accessibility to explore potential solutions. This approach not only provides a practical and scalable model, but also offers insights that can be generalized to larger-scale production in the future.

Conventionally, in-house PCB prototyping outsources various components, including raw board material, loose electronic components (ECs), engraved PCBs, and sometimes pre-assembled PCB assemblies (PCBA). Once the prototype is no longer useful, the PCB assemblies are usually discarded and contributes to e-waste, ending the short lifespan of the PCB prototypes and the materials used in them (gray path shown in Figure 1). To reduce the unsustainable impact caused by the continual procurement of materials and the generation of electronic waste, I propose the adoption of digital fabrication techniques that supplement existing processes and enable circular lifecycles for parts and materials used in PCB prototyping (shown in the green boundary in Figure 1). In the following sections, I highlight three research archetypes focus on achieving this: 1) a PCB assembly method that supports easy disassembly for EC recycling; 2) a reliable and temporary PCB substrate that allows easy material separation after disassembly, facilitating waste stream organization; 3) a PCB alteration method that renews and updates used PCBs to conserve raw substrate materials. Collectively, these approaches aim to establish a sustainable lifecycle for PCB prototyping materials, thereby reducing the demand for new materials, streamlining waste management processes, and mitigating overall

waste generation. These sustainable in-house PCB prototyping techniques have significant potential to be generalized and scaled up for industrial-level circuit manufacturing. All three methods align with the common manufacturing logic shared by existing in-house and industrial PCB production processes, facilitating broader application and impact.

2 RELATED WORK

Sustainable Interaction Design (SID) [3] highlights the importance of considering environmental impacts and human behaviors in designing interaction technologies. Subsequent work explored the impact of various design paradigms, particularly regarding the material outcomes of physical making [16, 20]. A large body of work has explored biobased or biodegradable materials to make prototypes of structures and mechanisms. Some took advantage of edible materials [1, 4] and materials derived from food waste [15], while others proposed design and fabrication methods of biobased materials, including dried microbial tissue such as bacteria [2] and fungi [8, 18], as well as elements harvested from living matter. Recent efforts have also demonstrated the potential impact of accessible fabrication methods in democratizing and advocating for sustainable actions. Zhu et al. [23] and Vasquez et al. [9] developed desktop machines to manufacture biobased and biodegradable e-textile fibers. Substiports retrofitted DFM 3D printing with Ad Hoc object insertion that allows scrap parts to be used as structural support to replace the newly extruded filament [19]. For PCB prototyping in particular, researchers in the HCI community have explored the role of sustainable materials to replace conductive and dielectric layer materials. For instance, Self-healing UI [13] introduced a composite material that self-repairs circuit wiring made with multi-wall carbon nanotubes, utilizing the dynamic cross-linking effect of polymer polyborosiloxane. Circuit Eraser [14] proposed using a regular eraser to remove circuit traces printed with conductive ink for fast circuit design iteration. Silver Tape [5] enables circuit trace repair via tape transfer of inkjet printed silver ink. Song et al. constructed back-yard decomposable electrical heater using edible golden leaf and a paper-based substrate [17]. In addition to material exploration, EcoEDA [11] explored software as part of electronic component sourcing infrastructure to support the reuse of electronic components from scraped electronic devices and apparatus.

3 DESIGNING SUSTAINABLE DIGITAL FABRICATION TECHNIQUES FOR PCB PROTOTYPING

In contrast to previous efforts in this field, I propose to supplement the PCB making process with additional phases that recover parts and materials from PCB assemblies at the end of their lifespan for reuse in future iterations and projects (green paths in Figure 1). Each enabled by a novel digital fabrication method, the following techniques open opportunities for PCBs to be designed for disassembly, material separation, and renewal.

3.1 Design for Disassembly

As shown in Figure 1, the conventional PCB prototyping process is concluded once the prototype has fulfilled its purpose of testing and

demonstration. A small portion of these prototypes is collected for recycling, but only if a suitable collection infrastructure exists for small-batch PCB recycling. Unfortunately, the majority end up in landfills, even though most of the electronic components soldered on the circuit board are still functional. When a new prototype is needed, new components are purchased for its purpose. The barrier to reusing the functional yet scraped components is the messiness, time consumption, and high level of skill required for prototypers to desolder components from the circuit board without causing damage. SolderlessPCB [21] is a novel and effective PCB assembly technique that uses housings to attach electronic components to circuit boards, making them easily detachable and reusable on-site, thereby conserving energy and resources otherwise spent on purchasing new components from manufacturers and reducing waste and environmental impact by minimizing discarded components. Specifically, a 3D housing model is generated based on the 3D model of the electronic components during the PCB design process. The housing can be fabricated using various desktop digital fabrication tools such as an SLA or FDM 3D printer, or a CNC milling machine. Users can assemble the PCB by sandwiching the electronic components between the board and the housing, securing the assembly with bolts and nuts, and disassembling the PCBA in the reverse order. Thread-based detachment does not require special skills to accomplish after the PCBA has reached the end of its lifespan, and all components can be reused in future iterations and projects. The solderless assembly method supports component recycling while accommodating various electrical functionalities (e.g. high-current application and high-frequency data transmission shown in Figure 2 f-i) with high reliability and conductivity performance.

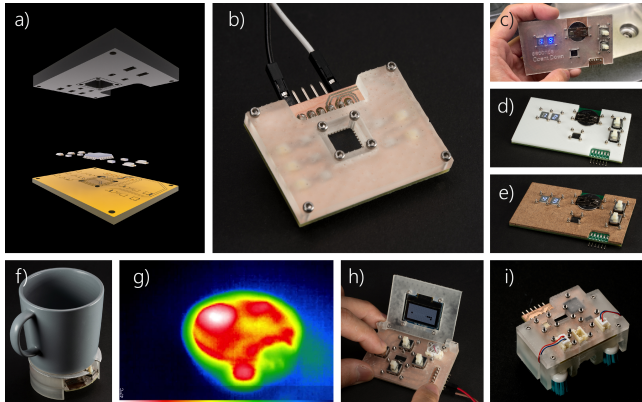


Figure 2: SolderlessPCB: a) rendering of an exploded view of a SolderlessPCB assembly, b) a photo of a SolderlessPCB assembly, c) a solderless kitchen timer made with FR4 and SLA 3D printed housing, d) and e) kitchen timer made with outsourced PCB, using FDM PLA and CNC MDF housings, f) a solderless mug heater, g) the heater reaches 47 Celsius, h) a solderless game console with I2C display, i) a solderless bristlebot equipped with ambient light sensor, LED and buzzer.

3.2 Design for Material Separation

At the downstream stage of disassembled PCBA, functional electronic components can be reused for future projects; however, custom PCBs are not easily reusable or recyclable. These boards contain conductors made of metal elements such as copper, gold, tin, lead, and the platinum group, as well as thermoset polymers and fiber-glass used in the dielectric layers. On an industrial scale, chemical and mechanical techniques are employed to reclaim these elements found in standard PCB waste [7, 12]. However, separating the metal and nonmetallic elements like resin and glue from the compound e-waste requires a tedious chemical process which posts dematerialization challenges and generates hazardous waste. Since PCB prototypes typically have a short lifespan due to their use in testing and demonstration, a temporary but reliable metal-nonmetal compound substrate that allows easy separation and high-resolution prototyping can significantly improve the recycling process without compromising the quality of the prototype.

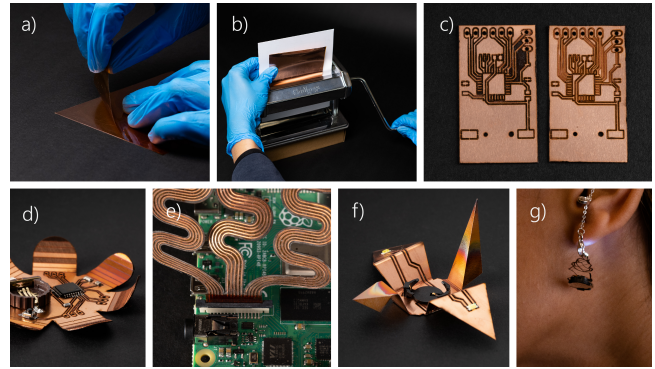


Figure 3: SolderlessPCB: a) hand lamination of copper foil and polyimide tape, b) remove air bubbles from composite, c) example of fiber circuit before and after isolation copper is removed, d) a fiber circuit arduino nano equivalent in a 2.5D shape, e) flexible camera cable made with fiber circuit, f) fiber circuit 3D crane, g) fiber circuit earring with shiny LED.

Fiber circuit [22] presents a fiber laser PCB prototyping technique that works with a copper-polyimide tape composite. Specifically, this composite consists of copper foil laminated with Kapton polyimide tape and can be easily assembled by hand (Figure 3 a and b). Circuit traces are engraved on the laminated substrate using a desktop fiber laser. The laser can be calibrated to selectively remove copper from the isolation areas between the traces (Figure 3 c). The silicone adhesive on the Kapton tape ensures satisfactory bonding, effectively holding the copper circuit traces in place.

Unlike FR4 or other conventional PCB composites that use UV-set polymers, the copper on Kapton tape can be easily removed when the circuit is no longer needed. This method facilitates the separation of materials, enabling the copper and polyimide to be disposed or recycled separately. By supporting material separation, Fiber circuit significantly enhances the recyclability of the waste interim prototype generated during the PCB development process, reducing the environmental impact.

3.3 Design for Renewal and Reuse

Beyond organizing the waste stream to support recycling PCB prototypes, renewing used circuit boards for new prototypes can further reduce material acquisition and disposal during multiple PCB design iterations and projects (shown as "renewal and update" in Figure 1). In traditional PCB manufacturing, circuit traces are created on the conductive layer of the substrate by removing material that does not correspond to any circuit traces. This permanent subtractive process binds the PCBs to specific schematics and functionalities, making them difficult to re-purpose for other applications.

An ongoing work, PCB Renewal, proposes a method to modify existing circuit traces using a combination of conductive epoxy deposition and CNC milling. This approach involves comparing the existing circuit trace patterns on a PCB with the desired updated ones and altering the circuit traces accordingly. Silver epoxy can be deposited in areas where copper was etched to form new conductors, and new traces can be engraved on the epoxy-filled copper surface. The epoxy deposition can be executed by hand with the guidance of a software visualization or automated by a custom desktop CNC milling machine equipped with a paste extruder.

When a PCB prototype is no longer useful, its components can be disassembled and cleaned for reuse, and the board itself can be renewed and updated for new designs in future iterations and projects. This method, when combined with SolderlessPCB, supports both effortless component reuse and in-house PCB renewal, collectively reviving conventionally discarded PCB elements and extending their lifespan.

4 FUTURE ENDEAVORS

Complementary to my PhD research direction, there are several other possibilities to be explored in achieving an environmental friendly PCB making process in the future.

4.1 Sustainable Material Compound for PCB Substrate

Previously in Fibercuit [22], I explored the possibility of constructing a temporary PCB substrate that is easily separable for waste stream management. However, traditional PCB substrates are still manufactured using hard-to-recycle materials such as fiberglass and polyimide [10]. To address this issue, an ongoing project is investigating the use of wood cellulose as the dielectric layer material for PCB substrates. Wood cellulose is an excellent choice for sustainability because it is a renewable resource that can be responsibly sourced, reducing the environmental impact from the outset. Additionally, wood cellulose is biodegradable, meaning it naturally decomposes without creating waste, thereby resolving sustainability challenges from both the raw material source and the end-of-life disposal. This innovative technique not only makes the PCB substrate biodegradable but also eliminates the need for industrial recycling processes, contributing to a more sustainable electronics industry.

4.2 Scaling and Deployment of Sustainable In-house PCB Prototyping Techniques

Future projects will also focus on the deployment of accessible fabrication methods for sustainable PCB prototyping. For decades, the maker community has played a central role in the development of digital fabrication, fostering open-source designs that democratize access to technology and innovation. Stakeholders in classrooms and communal makerspaces are often the primary users and practitioners of prototyping techniques, including PCB making. Recognizing their pivotal role, future efforts will aim to understand and address the practical challenges these communities face when adopting novel sustainable fabrication methods. This will involve qualitative studies, analysis, and discussions of detailed sustainability challenges and potential solutions, as well as iterative design and development of new technologies that support a circular lifecycle for PCB prototyping, ensuring less energy and resource consumption, little design and fabrication effort, and efficient material reuse and recycling.

5 CONCLUSION

By integrating sustainable practices and digital fabrication methods into the prototyping process from the ground up, I aim to create a more environmentally friendly and accessible approach to PCB fabrication, benefiting both the maker community and the broader electronics industry. I envision the technologies enabling the sustainable future of PCB manufacturing encompass, but are not limited to, the utilization of biobased and biodegradable materials, the reuse of electronic components, the support of material separation, and the renewal of scrap parts from e-waste. These innovations not only reduce the environmental impact of electronics manufacturing but also promote a circular economy, where resources are continually reused and recycled. Collectively the approaches address the entire lifecycle of PCBs, from material sourcing and manufacturing to disposal and recycling, fostering a more sustainable and responsible electronics prototyping community. Adopting these sustainable technologies and practices establishes a framework for future electronic innovation that is both environmentally responsible and technologically advanced.

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